

Digital Twin-Assisted Workload Allocation and Cooling Optimisation

Build report: status, findings, and data provenance

This report records the state of the research codebase, in the order the work happened, together with the origin of every input. It is written to be checked rather than trusted: each claim below is a number produced by a script in the repository and logged to *results/runs.csv* with its configuration hash, seed and git commit.

Phases complete	Tests	Simulated time	Logged runs	Commits
0, 0b, 1, 2, 3	144 passing	1,500 hours	173 rows	7 on main

1. Summary

The project set out to build a digital twin of a data hall whose internal state is a directed graph, train a message-passing network to predict per-rack inlet temperature, and drive two controllers from that forecast. Phases 0 through 3 are complete. Phases 4 (online calibration and drift), 5 (the two controllers) and 6 (the ablations) have not been started.

Two results reshaped the plan. The first is that the chosen simulator has no thermal physics at all, which forced its heat model to be rewritten before any comparison could be meaningful. The second is that on the resulting twin, neighbouring racks' temperatures correlate at 0.995, so spatial context adds far less to single-rack accuracy than the proposal assumed. The graph model still wins at every horizon, and it is the only model of the five that transfers to an unseen hall without refitting — which is where the paper's weight should now sit.

2. Chronology

Ten entries, in the order they occurred. The two marked as pivots changed the direction of the work.

#	Stage	Commit	Outcome	What happened
01	Setup	—	done	Machine is an Apple M2 carrying an x86_64 Homebrew and x86_64 Python. PyTorch has shipped no x86_64 macOS wheels since 2.2.2. Installed native arm64 uv (checksum-verified), pinned CPython 3.11.16 arm64, torch 2.14.0 with MPS.
02	Phase 0	8aba8b1	gate failed	SustainDC computes inlet temperature as a JSON constant plus the CRAC setpoint. Rack power never enters. A 14.8 kW swing on one rack moves every rack by exactly 0.000e+00 K. The influence matrix is the zero matrix, including its diagonal.
03	Decision	—	Option A	Keep SustainDC for its HVAC power chain and exogenous data; replace the inlet computation; state in the paper that the thermal ground truth is ours.
04	Phase 0b	00bb5a3	verified	Heat-recirculation kernel, $T = T_{\text{supply}} + D \cdot P$ with $D = K^{-1}[(I - A^T)^{-1} - I]$. Self-response 0.292 K, cross-rack up to 0.255 K, 29.7% of influence beyond one hop, setpoint gain 0.878–0.923 rather than pinned at 1.
05	Phase 1	6957b99	gate passed	1,500 h of trajectories across three halls at 30 s. Alibaba PAI jobs replayed, four placement policies, stepped cooling setpoints. Three calibration errors caught by measuring the output rather than trusting the config.
06	Phase 2	5e43bd6	finding	LightGBM with 2, 4 and 6 neighbours scores identically to three decimals. Own-rack features give $R^2 = 0.825$ on the 300 s target; adding every neighbour feature gives 0.827. Neighbour temperature is nearly redundant.
07	Correction	46545dd	task fixed	73.6% of the 300 s target variance came from setpoint changes that had not happened yet, against 4.3% from the state at t. The control trajectory became an input. LightGBM went from 0.653 to 0.188 K at 300 s.
08	Phase 3	d829879	complete	Graph model trained on hall_a only; hall_b and hall_c evaluated zero-shot with the same weights and the same normaliser. Wins at every horizon and holds its skill on unseen halls.

#	Stage	Commit	Outcome	What happened
09	Phase 2b	—	running	Baselines refit on hall_b and hall_c, so the transfer result has a bar to beat. Phase 2 covered hall_a only.
10	Phases 4–6	—	not started	Calibration and drift injection; the placement and cooling controllers; the five ablations.

3. The Phase 0 finding

Rack inlet temperature is computed in exactly one place in the whole SustainDC repository, *envs/datacenter.py:279*:

```
rack_inlet_temp = rack_supply_approach_temp + CRAC_setpoint
```

The approach temperature is a constant read from JSON at startup and never recomputed. No rack's power enters — not a neighbour's, not the rack's own. The consequence is that every model in the planned comparison would have tied at machine precision, and every ablation would have been null by construction. Two further blockers were found while probing: the gym wrapper hardcodes one scalar utilisation for all racks, so placement was inexpressible, and three of the four shipped hall configurations do not load.

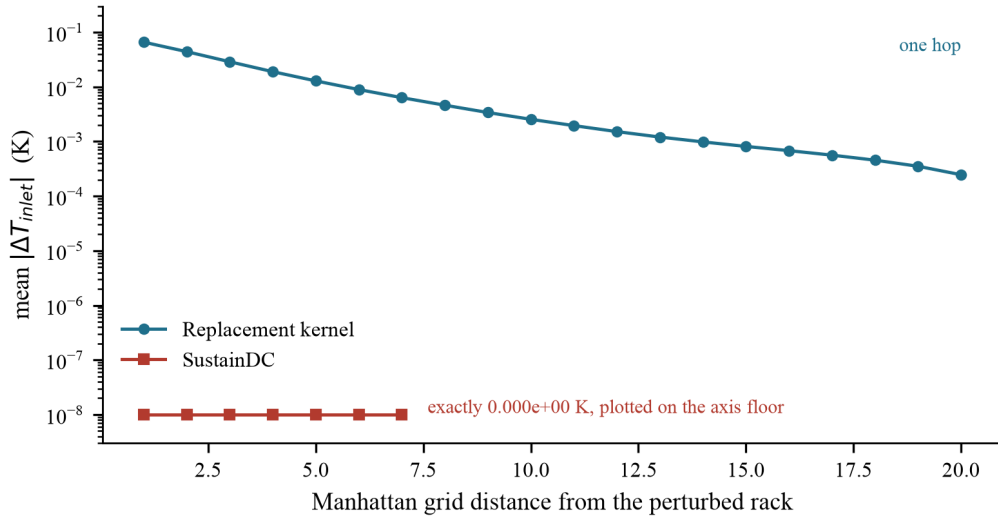


Figure 1. Response of every rack's inlet temperature to a 14.8 kW perturbation at one rack. SustainDC's values are exactly zero at every distance, including zero distance; its curve terminates at distance 7 because its default hall holds only 20 racks. The replacement kernel decays smoothly across the hall.

4. The replacement thermal model

With a_{ij} the fraction of rack i 's exhaust drawn into rack j 's inlet and K the diagonal matrix of rack thermal mass flows, the mixing energy balance at each inlet rearranges to $T = T_{supply} + D \cdot P$, with $D = K^{-1}[(I - A^T)^{-1} - I]$. The matrix inverse expands as a Neumann series in A^T , so multi-hop coupling emerges even though A is local: term m is air that passed through m racks before arriving. A is built from the floor plan alone — distance, rows crossed, and direction relative to the return-air path — never from the kernel the model is being asked to learn.

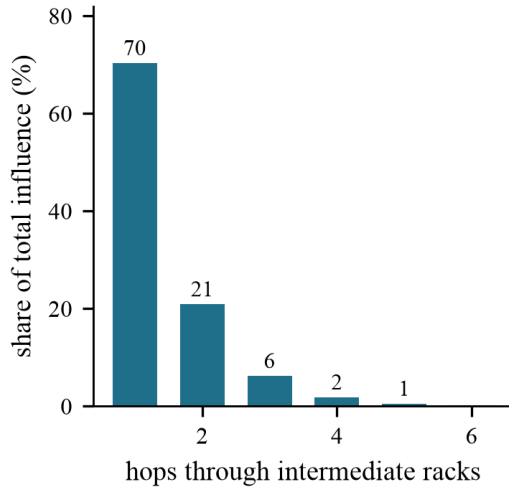


Figure 2. Share of total influence arriving after m hops. Roughly 30% arrives beyond the first, which is what justifies three message-passing layers.

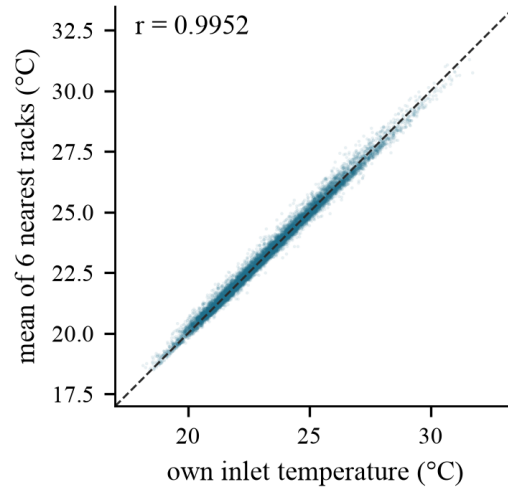


Figure 3. A rack's own inlet temperature against the mean of its six nearest racks, hall_a. The dashed line is equality.

The coupling is state-dependent, not a fixed matrix. K moves with rack fan speed, and the leakage fraction rises when the racks draw more air than the CRACs supply, so a cooling action reshapes the coupling rather than merely shifting the temperature field. SustainDC's setpoint gain was exactly 1.000000 at every rack, a rigid translation; the replacement's ranges from 0.878 to 0.923 and varies by rack.

A modelling bug worth recording

Normalising each source's leakage by geometry alone leaves a rack's recirculated intake fixed while its total intake grows with fan speed. Ramping its fans then dilutes its own inlet, and loading a rack from 50% to 100% utilisation *cooled* it by 0.79 K while its neighbours warmed by 0.08 K. A placement controller trained against that would have learned to stack load onto the hottest racks. Weighting each target's share of the recirculating air by its own thermal mass flow removes the artefact; two regression tests hold it removed.

5. The dataset

Real Alibaba PAI GPU jobs are replayed onto racks under four placement policies, including a deliberately bad corner-stacking policy so that the dataset contains hot states. Cooling setpoints are held for 5 to 20 minutes and then stepped, as a real CRAC controller behaves. 500 hours per hall at 30-second resolution, split by episode so that no input window crosses a boundary and every training timestep precedes every test timestep.

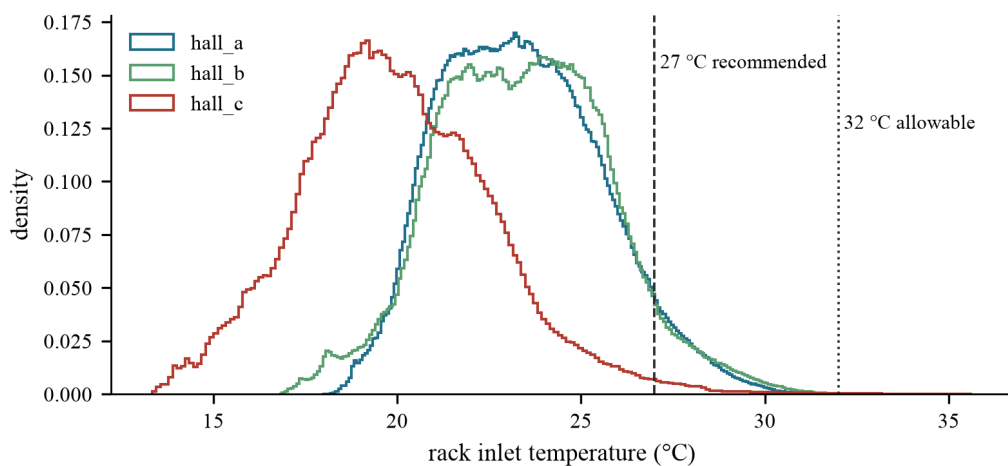


Figure 4. Distribution of rack inlet temperature for the three halls against the ASHRAE 2021 class A1 limits. The Phase 1 gate required at least 2% of timesteps to breach the recommended limit; the halls reach 18.9%, 18.8% and 19.3% respectively, and none exceeds the allowable limit materially. hall_c runs colder on average but reaches further, which is the signature of its uncontained layout.

6. Results

6.1 Accuracy on hall_a

Test RMSE of the temperature delta, in kelvin, over five seeds. The graph model wins at every horizon, with the largest margin at 60 s. That shape is consistent with the graph mattering most once heat has had time to propagate between racks but before the hall settles toward its steady state.

Model	Spatial input	30 s	60 s	300 s
Persistence	none	0.1057	0.1908	0.6367
RC network	graph edges	0.1682	0.1978	0.4833
Per-rack LSTM	none	0.0272	0.0598	0.2245
LightGBM $k=4$	4 nearest racks	0.0433	0.0606	0.1884
Graph model	message passing	0.0244	0.0434	0.1677

Margin over the best baseline: 11% at 30 s, 27% at 60 s, 11% at 300 s. Seed spread is ± 0.003 to ± 0.004 , so the ordering is stable.

6.2 Zero-shot transfer

The model is trained on hall_a and applied unchanged to the other two halls, carrying hall_a's normalisation statistics with it; those statistics are never recomputed on a target hall, which would be leakage. Results are given as skill, the fraction of the trivial predictor's error removed, because the three halls have different target variance and raw RMSE is not comparable across them.

Hall	Relationship to hall_a	30 s	60 s	300 s
hall_a	in-domain, 200 racks	76.0%	75.0%	70.8%
hall_b	140 racks, same aisle pattern	81.6%	78.9%	71.7%
hall_c	uncontained layout, moved CRACs	71.6%	74.4%	68.2%

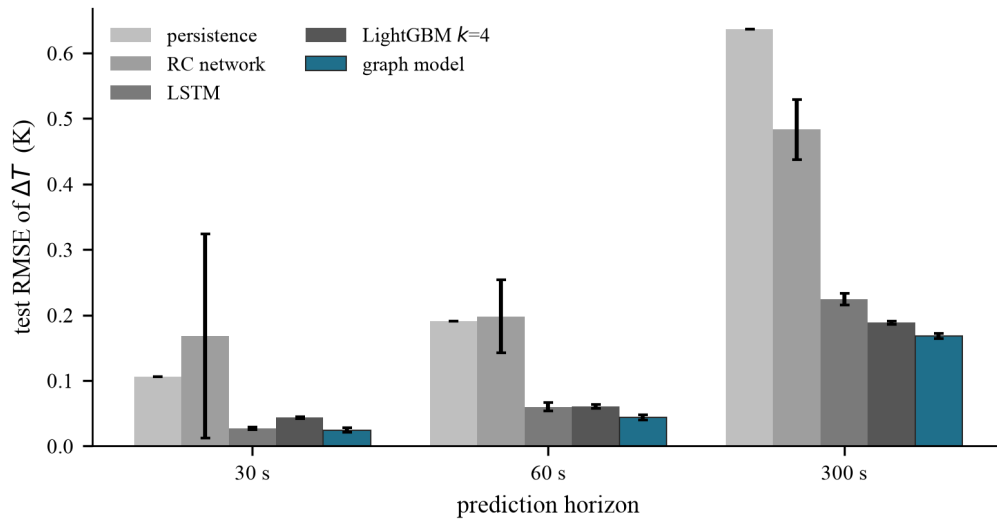


Figure 5. Accuracy on hall_a by prediction horizon. Error bars are one standard deviation over five seeds.

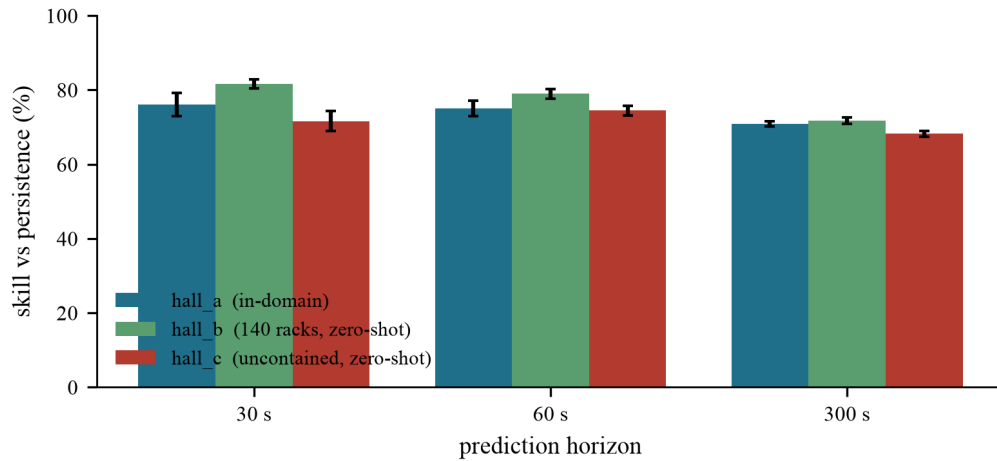


Figure 6. Skill against each hall's own persistence baseline. The transferred model removes as much of the trivial predictor's error on halls it has never seen as it does in-domain.

Transfer holds. On hall_b the transferred model removes more of persistence's error than it does in-domain, because that hall's larger target variance makes the trivial predictor worse. Layout transfer to hall_c costs roughly four skill points; scale transfer costs nothing. LightGBM's k -nearest features and the RC network's per-rack parameters cannot transfer at all, which is why this is the experiment with room in it.

7. Data provenance

After Phase 0, the inlet model, the thermal dynamics, the rack power curves and the hall geometry are all ours. What remains of SustainDC is its HVAC power chain and its exogenous data series. The paper cannot describe its thermal results as evaluated on SustainDC; the honest framing is that the twin is our own physics-based recirculation model, that SustainDC supplies the plant-side power chain, and that the thermal ground truth is therefore synthetic.

Component	Origin	Identifier
Rack inlet temperature	ours	src/twin/recirculation.py; formulation after Tang et al.
Thermal dynamics	ours	src/twin/thermal.py; first-order lag, $\tau = 60$ s
Rack power and airflow	ours	src/twin/power.py; SustainDC's curves rejected as hand-tuned
Hall geometry and aisles	ours	src/twin/geometry.py; SustainDC has no geometry
HVAC chain: chiller, tower, pumps	SustainDC	HewlettPackard/dc-rl @ a92b4755aca560e34a98d14028dda629eb968482
Weather and carbon intensity	SustainDC	same checkout, pinned by scripts/bootstrap_external.sh
Workload trace	Alibaba	alibaba/clusterdata, cluster-trace-gpu-v2020, pai_job_duration_estimate_100K.csv: 100,000 jobs over 289.8 h, 12.7 MB
Thermal limits	ASHRAE 2021	class A1: 27 °C recommended, 32 °C allowable
Trajectories	generated	data/trajectories/: hall_a 86 MB, hall_b 61 MB, hall_c 83 MB; regenerable from config and seed
All results	generated	results/runs.csv: 173 rows, each with config hash, seed and git SHA

The workload trace is bootstrapped rather than tiled: jobs are drawn independently from the empirical pool with arrivals following the trace's own hour-of-day profile. Tiling was rejected because the split is by time, so an exact repeat would place the same jobs in both training and test. It is a resampling of real data and should be described as such.

8. Open items before submission

#	Item	Why it matters
1	Re-run Phase 3 with memory headroom	The graph model's headline number came from a network of hidden width 32 with two attention heads, trained on 210,000 rows against LightGBM's 400,000. This machine has 8.6 GB of unified memory with several gigabytes already swapped; MPS measured about 70× slower than CPU once trajectories were resident, and activation memory hits a swap cliff that moves with the machine's state. The confound runs against the graph model, which won regardless, but it should not have to be explained away.
2	Baselines on hall_b and hall_c	Running at the time of writing. Without them the transfer table shows the graph model holding its skill on unseen halls with nothing measured on those halls to compare against.
3	Phase 6 before Phase 5	The edge-shuffle ablation directly tests the Phase 2 prediction that the graph is close to decorative. It is cheap, and it is the result most likely to change how the paper is framed. Building two controllers on top of a graph that carries little information would be an expensive way to discover this.
4	State the kernel limitation plainly	The 0.995 neighbour correlation is a property of our recirculation kernel's decay length. A hall with tighter, more local recirculation would show more spatial signal and give a graph model more room. This belongs in the limitations section.